

Maranello AI - Enterprise Knowledge Base

Quality & Manufacturing Operations

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Consolidated version of the five fictional internal documents used by the Maranello AI RAG system.

Created exclusively for academic demonstration purposes.
This project is fictional and is not affiliated with Ferrari N.V. or any real automotive manufacturer.

1 Manufacturing Quality Policy

Document ID: MQP-001

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Classification: Internal

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Review Cycle: Annual

Status: Active

1.1 1. Purpose

This policy defines the quality standards, performance thresholds, monitoring requirements, and escalation criteria applicable to manufacturing operations.

The purpose of the policy is to ensure that production quality is continuously monitored and that deviations from expected performance are identified, investigated, and addressed consistently.

This policy applies to all production lines, shifts, vehicle models, manufacturing teams, and production batches included within the Quality & Manufacturing Operations scope.

1.2 2. Quality Principles

Manufacturing operations shall follow the following principles:

1. product quality takes priority over production volume;
 2. quality deviations must be identified as early as possible;
 3. production data must be reviewed using consistent quality indicators;
 4. significant deviations require documented investigation;
 5. repeated quality issues require escalation even when individual observations remain below critical thresholds;
 6. decisions affecting product conformity must be traceable;
 7. production shall not continue without review when critical quality conditions are identified.
-

1.3 3. Core Quality Indicators

The following indicators are used to monitor manufacturing quality:

- Defect Rate;
- Rework Rate;
- Scrap Rate;
- Quality Score;
- production-line downtime;
- recurring defect categories;
- supplier-related quality performance.

These indicators may be evaluated at batch, production-line, shift, vehicle-model, supplier, or time-period level.

1.4 4. Defect Rate

1.4.1 4.1 Definition

The Defect Rate represents the percentage of produced units identified as defective.

It is calculated as:

$$\text{Defect Rate (\%)} = (\text{defective_units} / \text{units_produced}) \times 100$$

1.4.2 4.2 Defect Rate Thresholds

The following thresholds apply:

Defect Rate	Classification	Required Action
<= 2.0%	Normal	Continue standard monitoring.
> 2.0% and <= 3.5%	Warning	Perform quality review and monitor subsequent production.
> 3.5%	Critical	Initiate formal escalation and non-conformity assessment.

A Critical classification does not automatically require permanent production shutdown.

The responsible Quality team must evaluate the affected batch, defect severity, recurrence, and potential customer impact before determining the appropriate operational response.

1.5 5. Rework Rate

1.5.1 5.1 Definition

The Rework Rate represents the percentage of produced units requiring corrective manufacturing operations before they can be accepted.

It is calculated as:

$$\text{Rework Rate (\%)} = (\text{rework_units} / \text{units_produced}) \times 100$$

1.5.2 5.2 Rework Thresholds

Rework Rate	Classification	Required Action
<= 1.5%	Normal	Standard monitoring.
> 1.5% and <= 3.0%	Warning	Review the dominant defect category and production conditions.
> 3.0%	Critical	Initiate root-cause analysis and quality escalation.

Repeated Warning classifications for the same production line during three consecutive monitored periods must be treated as a Critical condition.

1.6 6. Scrap Rate

1.6.1 6.1 Definition

The Scrap Rate represents the percentage of produced units that cannot be recovered through approved rework activities.

It is calculated as:

Scrap Rate (%) =
(scrap_units / units_produced) × 100

1.6.2 6.2 Scrap Thresholds

Scrap Rate	Classification	Required Action
<= 0.5%	Normal	Standard monitoring.
> 0.5% and <= 1.0%	Warning	Review scrap causes and affected components.
> 1.0%	Critical	Initiate quality escalation and documented investigation.

Scrap associated with safety-critical components must be reviewed regardless of the overall Scrap Rate.

1.7 7. Quality Score

1.7.1 7.1 Definition

The Quality Score is an internal manufacturing quality indicator expressed on a scale from 0 to 100. Higher values represent stronger overall quality performance.

1.7.2 7.2 Quality Score Thresholds

Quality Score	Classification	Required Action
>= 95	Target	No corrective action required.
>= 90 and < 95	Acceptable	Continue monitoring.
>= 85 and < 90	Warning	Perform quality review.
< 85	Critical	Initiate formal investigation and escalation.

A missing or invalid Quality Score must not be interpreted as acceptable performance.

The underlying data must be validated before a quality decision is made.

1.8 8. Production-Line Downtime

Production downtime is monitored as both an operational and quality-related indicator.

The following thresholds apply to downtime associated with a production batch:

Downtime	Classification	Required Action
<= 45 minutes	Normal	Standard monitoring.
> 45 and <= 90 minutes	Warning	Review operational cause.
> 90 minutes	Critical	Escalate to Manufacturing Operations and investigate the cause.

Downtime greater than 180 minutes requires immediate management notification.

Extreme or implausible downtime values must first be validated to exclude data-quality issues before operational conclusions are drawn.

1.9 9. Recurring Quality Issues

A quality issue must be considered recurring when the same significant defect pattern appears repeatedly within the same:

- production line;
- supplier;
- component category;
- vehicle model;
- or production shift.

Repeated Warning conditions may require escalation even if no individual observation exceeds a Critical threshold. The Quality team must evaluate both the severity and frequency of the observed issue.

1.10 10. Supplier-Related Quality Monitoring

Supplier performance must be monitored when manufacturing defects can reasonably be associated with externally supplied components.

Supplier-related analysis should consider:

- defect frequency;
- defect rate;
- affected component category;
- recurrence over time;
- scrap generated;
- rework generated;
- severity of the detected defects.

A supplier must not be classified as non-compliant solely on the basis of one isolated batch unless the issue involves a safety-critical or otherwise severe defect.

Detailed supplier escalation requirements are defined in the Supplier Quality Procedure.

1.11 11. Data Quality Requirements

Quality decisions must be based on validated production data.

Before calculating or interpreting manufacturing KPIs, analysts and automated systems must evaluate relevant data for:

- missing values;
- duplicate records;
- inconsistent categories;
- invalid numerical ranges;
- inconsistent date formats;
- abnormal outliers;
- logical inconsistencies between related fields.

Examples of logically invalid records include:

```
defective_units > units_produced  
quality_score > 100
```

Records containing significant data-quality issues must be corrected, excluded, or explicitly flagged before being used for operational decisions.

Automated cleaning must not silently modify data when the correct value cannot be determined with reasonable confidence.

1.12 12. Quality Escalation

A formal quality escalation must be initiated when one or more of the following conditions occur:

- Defect Rate exceeds 3.5%;
- Rework Rate exceeds 3.0%;
- Scrap Rate exceeds 1.0%;
- Quality Score is below 85;
- production-line downtime exceeds 90 minutes;
- repeated Warning conditions indicate a persistent quality problem;
- a safety-critical defect is identified;
- a significant supplier-related quality issue is detected.

The escalation must identify:

1. the affected production scope;
2. the relevant KPI or quality issue;
3. available supporting evidence;
4. immediate containment actions;
5. responsible stakeholders;
6. required follow-up activities.

Detailed operational escalation rules are defined in the Production Escalation Policy.

1.13 13. Responsibilities

1.13.1 Quality Engineer

Responsible for:

- reviewing quality deviations;
- validating quality indicators;
- initiating non-conformity assessments;
- supporting root-cause analysis;
- documenting quality decisions.

1.13.2 Manufacturing Supervisor

Responsible for:

- monitoring production conditions;
- coordinating operational containment;
- supporting investigations;
- implementing approved corrective actions.

1.13.3 Supplier Quality Engineer

Responsible for:

- investigating supplier-related issues;
- evaluating supplier performance;
- coordinating supplier corrective actions;
- escalating recurring supplier quality problems.

1.13.4 Quality Manager

Responsible for:

- reviewing Critical quality conditions;
 - approving major quality decisions;
 - coordinating cross-functional escalations;
 - ensuring compliance with internal quality procedures.
-

1.14 14. Use of Automated Analytical Systems

Automated analytical systems, including Maranello AI, may support quality monitoring and decision-making by:

- calculating manufacturing KPIs;
- identifying trends and anomalies;
- retrieving applicable internal policies;
- highlighting conditions requiring review;
- generating analytical summaries.

Automated systems provide decision support and do not replace required human approval for critical manufacturing or quality decisions.

When data quality is insufficient, the system should explicitly communicate the limitation rather than present uncertain results as verified facts.

1.15 15. Related Documents

This policy should be interpreted together with:

- Non-Conformity Management Procedure;
 - Supplier Quality Procedure;
 - Rework and Scrap Procedure;
 - Production Escalation Policy.
-

1.16 16. Document Control

Field	Value
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2 Non-Conformity Management Procedure

Document ID: NCP-001

Version: 1.0

Owner: Quality & Manufacturing Operations

Classification: Internal

Effective Date: 2025-01-01

Review Cycle: Annual

Status: Active

2.1 1. Purpose

This procedure defines the process for identifying, containing, investigating, managing, and closing manufacturing non-conformities.

The objective is to ensure that quality deviations are handled consistently, affected production is appropriately controlled, root causes are investigated, and corrective actions are documented and verified.

2.2 2. Scope

This procedure applies to non-conformities associated with:

- production batches;
- production lines;
- manufactured components;
- supplier-provided components;
- rework activities;
- inspection results;
- manufacturing processes;
- quality KPI deviations.

The procedure applies to all production lines and manufacturing teams within the Quality & Manufacturing Operations scope.

2.3 3. Definition of Non-Conformity

A non-conformity is any condition in which a product, component, process, production batch, or manufacturing result fails to satisfy an applicable quality requirement.

Examples include:

- excessive Defect Rate;
- excessive Rework Rate;
- excessive Scrap Rate;
- Quality Score below the required threshold;
- failed quality inspection;
- recurring defect patterns;
- safety-critical defects;
- significant supplier-related defects;
- production process deviations.

A data-quality anomaly alone does not automatically represent a manufacturing non-conformity.

The underlying production data must first be validated.

2.4 4. Non-Conformity Identification

A potential non-conformity may be identified through:

- routine quality inspection;
- automated KPI monitoring;
- manufacturing operator observation;
- supplier quality inspection;
- production-line monitoring;
- trend analysis;
- audit activities;
- automated analytical systems.

When a potential non-conformity is identified, the available evidence must be reviewed before a final classification is assigned.

2.5 5. Initial Data Validation

Before initiating a formal non-conformity based on analytical data, the responsible team must verify that the supporting data is sufficiently reliable.

The validation should consider:

- missing values;
- duplicate observations;
- invalid numerical values;
- inconsistent categorical values;
- incorrect date formats;
- abnormal outliers;
- logical inconsistencies.

For example:

```
defective_units > units_produced
```

must be treated as a data-quality problem until the correct production values are verified.

Automated systems must not interpret obviously invalid data as confirmed manufacturing performance.

2.6 6. Non-Conformity Classification

Non-conformities are classified according to severity.

2.6.1 Minor

A Minor non-conformity:

- has limited operational impact;
- does not affect safety-critical characteristics;
- can be contained locally;
- does not indicate a recurring systemic problem.

Typical response:

- document the issue;
- perform local correction;
- continue monitoring.

2.6.2 Major

A Major non-conformity:

- exceeds an established quality threshold;
- affects multiple units or batches;
- requires structured investigation;
- may indicate a recurring manufacturing or supplier issue.

Typical response:

- initiate containment;
- assign investigation responsibility;
- perform root-cause analysis;
- define corrective actions.

2.6.3 Critical

A Critical non-conformity:

- may affect product safety;
- presents significant customer risk;
- indicates severe process loss of control;
- involves repeated unresolved Major issues;
- requires immediate management attention.

Typical response:

- immediate containment;
 - production escalation;
 - management notification;
 - formal investigation;
 - documented authorization before normal production resumes.
-

2.7 7. Immediate Containment

When a Major or Critical non-conformity is confirmed, containment activities must begin without unnecessary delay.

Possible containment actions include:

- isolating the affected batch;
- preventing shipment or further processing;
- increasing inspection frequency;
- performing additional quality checks;
- temporarily holding related batches;
- separating suspected components;
- notifying relevant manufacturing personnel.

Containment is intended to control immediate risk.

It does not replace root-cause analysis or corrective action.

2.8 8. Batch Isolation

Affected production batches must remain traceable throughout the investigation.

The containment record should identify:

- batch identifier;

- production date;
- production line;
- vehicle model;
- shift;
- relevant defect category;
- affected supplier or component where applicable;
- inspection status;
- reason for isolation.

A batch must not be released solely because subsequent production appears normal.

Release requires appropriate quality verification.

2.9 9. Root-Cause Analysis

Major and Critical non-conformities require investigation of the underlying cause.

Root-cause analysis may consider:

- production equipment;
- manufacturing methods;
- materials;
- supplier components;
- operator activities;
- environmental conditions;
- process parameters;
- inspection methods;
- recurring historical patterns.

Appropriate investigation techniques may include:

- 5 Whys;
- Ishikawa analysis;
- Pareto analysis;
- trend analysis;
- comparison between production lines;
- comparison between shifts;
- supplier performance analysis.

The investigation must distinguish correlation from confirmed causation.

2.10 10. Corrective Actions

Corrective actions must address the identified or most probable root cause.

Examples include:

- production parameter adjustment;
- equipment maintenance;
- operator instruction;
- additional inspection;
- supplier corrective action;
- process redesign;
- component replacement;
- revised quality control.

Each corrective action should identify:

1. the action to be performed;
 2. the responsible person or team;
 3. the expected completion date;
 4. the verification method;
 5. the resulting status.
-

2.11 11. Verification of Effectiveness

A corrective action cannot be considered complete solely because it has been implemented.

Its effectiveness must be verified.

Verification may include:

- monitoring subsequent batches;
- comparing defect rates before and after the action;
- reviewing Quality Score trends;
- checking recurrence of the defect category;
- verifying supplier performance;
- performing additional inspections.

If the problem persists, the investigation must be reopened or escalated.

2.12 12. Recurring Non-Conformities

Repeated quality deviations must be evaluated collectively rather than as isolated events.

A recurring condition may exist when:

- the same defect category repeatedly affects the same production line;
- the same supplier is repeatedly associated with quality deviations;
- the same shift consistently exhibits abnormal performance;
- corrective actions fail to prevent recurrence;
- Warning conditions continue across multiple monitoring periods.

Repeated Minor or Warning conditions may therefore justify Major classification or formal escalation.

2.13 13. Supplier-Related Non-Conformities

When evidence indicates that a non-conformity may be associated with a supplied component, the Supplier Quality Engineer must be involved.

The investigation should consider:

- supplier identifier;
- component category;
- affected batches;
- recurrence;
- defect severity;
- scrap and rework impact.

Supplier-related corrective actions must follow the Supplier Quality Procedure.

2.14 14. Rework and Scrap

When affected units may be recovered through rework, all activities must follow the Rework and Scrap Procedure. Reworked units must be inspected again before being accepted.

Units that cannot be restored to applicable quality requirements must be evaluated for scrap according to the applicable procedure.

2.15 15. Escalation Criteria

A non-conformity must be escalated when:

- a Critical quality condition is identified;
- a safety-critical defect is detected;
- Defect Rate exceeds the Critical threshold defined by the Manufacturing Quality Policy;
- Rework Rate exceeds the Critical threshold;
- Scrap Rate exceeds the Critical threshold;
- Quality Score falls below the Critical threshold;
- the same significant issue repeatedly occurs;
- corrective actions are ineffective;
- a significant supplier-related issue is identified.

Operational escalation must follow the Production Escalation Policy.

2.16 16. Non-Conformity Closure

A non-conformity may be closed only when:

1. the affected production scope has been identified;
2. containment has been completed where required;
3. the issue has been investigated;
4. required corrective actions have been implemented;
5. effectiveness has been verified;
6. required documentation has been completed;
7. no unresolved Critical risk remains.

Closure must be traceable and approved by the appropriate Quality authority.

2.17 17. Responsibilities

2.17.1 Quality Engineer

Responsible for:

- validating the quality issue;
- classifying the non-conformity;
- coordinating investigation;
- documenting findings;
- verifying corrective actions.

2.17.2 Manufacturing Supervisor

Responsible for:

- implementing containment;
- supporting investigation;

- coordinating production activities;
- implementing approved operational corrections.

2.17.3 Supplier Quality Engineer

Responsible for:

- managing supplier-related investigations;
- coordinating supplier corrective actions;
- monitoring supplier performance.

2.17.4 Quality Manager

Responsible for:

- reviewing Critical non-conformities;
- approving significant quality decisions;
- supporting cross-functional escalation;
- approving closure where required.

2.18 18. Use of Maranello AI

Maranello AI may support non-conformity management by:

- detecting KPI deviations;
- identifying abnormal manufacturing patterns;
- analyzing historical production data;
- retrieving applicable quality procedures;
- identifying potentially recurring issues;
- supporting investigation with quantitative evidence.

Maranello AI provides decision support.

It must not autonomously approve:

- release of isolated production batches;
- closure of Major or Critical non-conformities;
- safety-critical decisions;
- final root-cause conclusions where human validation is required.

When supporting data is incomplete or unreliable, the system should clearly indicate the limitation.

2.19 19. Related Documents

This procedure should be interpreted together with:

- Manufacturing Quality Policy;
- Supplier Quality Procedure;
- Rework and Scrap Procedure;
- Production Escalation Policy.

2.20 20. Document Control

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3 Supplier Quality Procedure

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Classification: Internal

Effective Date: 2025-01-01

Review Cycle: Annual

Status: Active

3.1 1. Purpose

This procedure defines the requirements for monitoring, evaluating, investigating, and escalating quality issues associated with externally supplied components used in manufacturing operations.

The objective is to ensure that supplier-related quality deviations are identified using reliable evidence and managed consistently according to their severity, recurrence, and operational impact.

3.2 2. Scope

This procedure applies to suppliers and supplied components associated with production batches within the Quality & Manufacturing Operations scope.

Supplier quality evaluation may consider:

- manufacturing defects;
 - component-related non-conformities;
 - rework;
 - scrap;
 - recurring defect patterns;
 - inspection failures;
 - production disruption associated with supplied components.
-

3.3 3. Supplier Quality Principles

Supplier quality management shall follow these principles:

1. supplier performance must be evaluated using validated data;
 2. isolated observations should not automatically result in supplier escalation;
 3. recurring quality problems require investigation;
 4. defect severity must be considered together with defect frequency;
 5. safety-critical issues require immediate review regardless of historical performance;
 6. supplier-related conclusions must distinguish correlation from confirmed causation;
 7. corrective actions must be documented and their effectiveness verified.
-

3.4 4. Supplier Performance Indicators

Supplier quality performance should be evaluated using available manufacturing indicators.

Relevant indicators include:

- Defect Rate;
- defective units;

- Rework Rate;
- Scrap Rate;
- recurring defect categories;
- affected component categories;
- failed inspections;
- number of affected production batches;
- persistence of quality issues over time.

No individual KPI should be interpreted without considering the production context and quality of the underlying data.

3.5 5. Supplier Defect Rate

3.5.1 5.1 Definition

Supplier Defect Rate represents the percentage of produced units identified as defective within production batches associated with a specific supplier.

It is calculated as:

$$\text{Supplier Defect Rate (\%)} = \frac{\text{(total defective units associated with supplier / total units produced in associated batches)} \times 100$$

3.5.2 5.2 Performance Thresholds

The following thresholds are used for supplier quality monitoring:

Supplier Defect Rate	Classification	Required Action
<= 2.0%	Normal	Continue standard monitoring.
> 2.0% and <= 3.0%	Observation	Review supplier trend and affected component categories.
> 3.0% and <= 4.0%	Warning	Initiate formal supplier quality review.
> 4.0%	Critical	Initiate supplier escalation and corrective-action process.

These thresholds apply to aggregated supplier performance over a meaningful production scope.

An isolated production batch must not automatically determine the overall supplier classification.

3.6 6. Minimum Evidence for Supplier Evaluation

Supplier performance should not be classified using an insufficient sample.

Before assigning a Warning or Critical classification based on aggregated performance, the Supplier Quality Engineer should consider:

- number of associated batches;
- total production volume;
- affected component categories;
- observation period;
- recurrence of defects.

When the available evidence is insufficient, the result should be reported as requiring additional monitoring rather than as a confirmed supplier performance issue.

3.7 7. Recurring Supplier Issues

A supplier-related issue should be considered recurring when similar quality deviations are observed repeatedly across:

- multiple production batches;
- multiple production dates;
- the same component category;
- different production lines using the same supplied component.

Three or more significant supplier-related quality events within a rolling 30-day period require formal supplier quality review.

Two consecutive monitored periods classified as Warning must also trigger formal supplier review.

3.8 8. Critical Supplier Events

Immediate supplier quality escalation is required when:

- Supplier Defect Rate exceeds 4.0% over a validated monitoring period;
- a safety-critical supplied component fails inspection;
- the same Major non-conformity repeatedly occurs after corrective action;
- supplier-related defects cause significant production disruption;
- supplier-related defects create substantial scrap or rework;
- evidence indicates loss of control in the supplier manufacturing process.

Critical supplier events must be communicated to the Quality Manager and Supplier Quality Engineer.

3.9 9. Supplier Quality Review

A formal supplier quality review should evaluate:

1. the supplier identifier;
2. affected component categories;
3. affected production batches;
4. production volume;
5. defect frequency;
6. defect categories;
7. Defect Rate;
8. Rework Rate;
9. Scrap Rate;
10. historical recurrence;
11. operational impact;
12. available evidence regarding root cause.

The review must distinguish between statistical association and confirmed supplier responsibility.

3.10 10. Supplier Corrective Action Request

When a supplier-related Major or Critical issue is confirmed, a Supplier Corrective Action Request may be initiated.

The request should include:

- description of the problem;
- affected component;
- available supporting evidence;
- affected production scope;
- containment requirements;
- requested root-cause analysis;
- required corrective action;
- response deadline.

Unless otherwise determined by the Supplier Quality Engineer, the supplier should provide an initial containment response within **2 business days** and a documented corrective-action plan within **10 business days**.

3.11 11. Containment Requirements

Supplier-related containment may include:

- isolation of suspected components;
- increased incoming inspection;
- additional production inspection;
- temporary segregation of supplier inventory;
- verification of related batches;
- controlled replacement of affected components.

Containment requirements must be proportional to the severity and potential impact of the issue.

3.12 12. Verification of Corrective Actions

Supplier corrective actions must be verified before the issue is considered resolved.

Verification may include:

- monitoring subsequent batches;
- comparing supplier Defect Rate before and after corrective action;
- reviewing recurrence of the affected defect category;
- evaluating Rework Rate;
- evaluating Scrap Rate;
- performing additional component inspection.

If performance does not improve, the issue must remain open and may require further escalation.

3.13 13. Supplier Performance Recovery

A supplier previously classified as Warning or Critical may return to standard monitoring when:

- required corrective actions have been implemented;
- effectiveness has been verified;
- no unresolved Critical non-conformity remains;
- subsequent production demonstrates acceptable performance;
- the Supplier Quality Engineer approves return to standard monitoring.

A single compliant batch is not sufficient evidence of sustained recovery following a Critical supplier issue.

3.14 14. Data Quality Requirements

Supplier analysis must use validated manufacturing data.

Before calculating supplier performance, the analysis must account for:

- missing supplier identifiers;
- duplicated production records;
- inconsistent supplier identifier formatting;
- invalid production quantities;
- inconsistent component categories;
- abnormal numerical values.

For example:

SUP-07

SUP-07

must be normalized to the same supplier identifier before aggregated performance is calculated.

Records with missing supplier identifiers must not be automatically assigned to a supplier.

3.15 15. Supplier Comparison

When comparing suppliers, the analysis should consider differences in:

- production volume;
- number of associated batches;
- component category;
- production period;
- defect severity.

Ranking suppliers solely by total defective units may produce misleading conclusions when suppliers are associated with substantially different production volumes.

Rate-based metrics should therefore normally be preferred for comparative quality analysis.

3.16 16. Relationship with Non-Conformity Management

When a supplier-related quality issue results in a confirmed manufacturing non-conformity, the Non-Conformity Management Procedure must also be applied.

Supplier quality management does not replace internal containment responsibilities.

The manufacturing organization remains responsible for controlling affected production while supplier investigation is in progress.

3.17 17. Responsibilities

3.17.1 Supplier Quality Engineer

Responsible for:

- monitoring supplier quality performance;
- evaluating supplier-related deviations;
- initiating supplier reviews;
- coordinating corrective actions;
- verifying effectiveness;

- approving return to standard monitoring.

3.17.2 Quality Engineer

Responsible for:

- providing manufacturing quality evidence;
- identifying affected batches;
- supporting non-conformity classification;
- supporting root-cause analysis.

3.17.3 Manufacturing Supervisor

Responsible for:

- implementing required operational containment;
- supporting traceability;
- coordinating affected production activities.

3.17.4 Quality Manager

Responsible for:

- reviewing Critical supplier issues;
- approving major escalations;
- coordinating cross-functional decisions where required.

3.18 18. Use of Maranello AI

Maranello AI may support supplier quality activities by:

- calculating supplier-specific Defect Rates;
- comparing supplier performance;
- identifying recurring defect patterns;
- analyzing affected component categories;
- identifying suppliers associated with abnormal rework or scrap;
- retrieving applicable supplier quality rules;
- highlighting suppliers requiring review according to documented thresholds.

For example, Maranello AI may determine that a supplier has a Defect Rate above the applicable Warning threshold and retrieve this procedure to explain the required next action.

The system must clearly distinguish between:

Statistical association

and:

Confirmed supplier root cause

Maranello AI must not claim that a supplier caused a defect solely because the supplier is statistically associated with affected batches.

3.19 19. Related Documents

This procedure should be interpreted together with:

- Manufacturing Quality Policy;
- Non-Conformity Management Procedure;

- Rework and Scrap Procedure;
- Production Escalation Policy.

3.20 20. Document Control

Field	Value
Document ID	SQP-001
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Owner	Supplier Quality & Manufacturing Operations
Classification	Internal
Effective Date	2025-01-01
Review Cycle	Annual
Status	Active

4 Rework and Scrap Procedure

Document ID: RSP-001

Version: 1.0

Owner: Quality & Manufacturing Operations

Classification: Internal

Effective Date: 2025-01-01

Review Cycle: Annual

Status: Active

4.1 1. Purpose

This procedure defines the requirements for evaluating, authorizing, performing, verifying, and documenting rework and scrap activities associated with manufacturing quality deviations.

The objective is to ensure that non-conforming units are managed consistently, that reworked products satisfy applicable quality requirements before release, and that scrap decisions remain controlled and traceable.

4.2 2. Scope

This procedure applies to:

- production batches;
- defective manufactured units;
- components requiring corrective processing;
- units undergoing rework;
- units classified as scrap;
- supplier-related defective components;
- quality deviations identified during inspection.

The procedure applies to all manufacturing operations within the Quality & Manufacturing Operations scope.

4.3 3. Definitions

4.3.1 Rework

Rework is an approved manufacturing activity performed on a non-conforming unit with the objective of restoring it to applicable quality requirements.

A reworked unit must satisfy the same applicable acceptance requirements as a unit that did not require rework.

4.3.2 Scrap

Scrap is the controlled disposition of a unit or component that cannot be economically, technically, or safely restored to applicable quality requirements through an approved rework process.

Scrapped units must not return to normal production flow.

4.4 4. Rework Principles

Rework activities shall follow these principles:

1. rework must be technically appropriate for the identified defect;
2. rework must not introduce additional uncontrolled quality risks;

3. affected units must remain traceable;
4. reworked units require verification before acceptance;
5. repeated rework patterns require investigation;
6. rework must not be used to conceal recurring manufacturing problems;
7. safety-critical characteristics require appropriate quality approval.

4.5 5. Rework Rate

4.5.1 5.1 Definition

The Rework Rate represents the percentage of produced units requiring approved corrective manufacturing operations.

It is calculated as:

$$\text{Rework Rate (\%)} = (\text{rework_units} / \text{units_produced}) \times 100$$

4.5.2 5.2 Rework Rate Thresholds

The following thresholds apply:

Rework Rate	Classification	Required Action
<= 1.5%	Normal	Continue standard monitoring.
> 1.5% and <= 3.0%	Warning	Review defect categories and production conditions.
> 3.0%	Critical	Initiate root-cause analysis and quality escalation.

These thresholds are consistent with the Manufacturing Quality Policy.

Three consecutive monitored periods classified as Warning for the same production line must be treated as a Critical condition.

4.6 6. Rework Authorization

Rework may be performed only when:

- the affected defect has been identified;
- an appropriate corrective operation exists;
- the rework activity does not create unacceptable additional risk;
- traceability can be maintained;
- required quality verification can be performed.

Rework involving safety-critical characteristics requires explicit Quality approval before the affected unit can be released.

4.7 7. Rework Execution

The rework process should identify:

- affected batch;
- affected unit or component where applicable;
- defect category;

- required corrective activity;
- responsible manufacturing team;
- date of rework;
- verification requirements.

Rework must be performed using approved manufacturing or quality instructions.

Uncontrolled or undocumented rework is not permitted.

4.8 8. Post-Rework Verification

Every reworked unit must undergo appropriate verification before acceptance.

Verification may include:

- visual inspection;
- dimensional inspection;
- functional testing;
- electrical testing;
- process verification;
- additional quality checks appropriate to the defect.

A reworked unit must not be considered acceptable solely because the corrective activity was completed.

Acceptance requires successful verification.

4.9 9. Repeated Rework

Repeated rework associated with the same:

- production line;
- defect category;
- vehicle model;
- component category;
- supplier;
- or production shift

must be evaluated as a potential systemic quality issue.

High rework volume may indicate that the underlying production process is not operating under adequate control.

Correcting individual units does not eliminate the requirement to investigate recurring causes.

4.10 10. Scrap Principles

A unit or component should be evaluated for scrap when:

- approved rework cannot restore conformity;
- rework would introduce unacceptable quality risk;
- repeated rework has failed;
- the component has suffered irreversible damage;
- applicable quality requirements cannot be verified;
- continued use would create a safety or reliability concern.

Scrap decisions must remain traceable.

4.11 11. Scrap Rate

4.11.1 11.1 Definition

The Scrap Rate represents the percentage of produced units permanently rejected from normal production flow. It is calculated as:

$$\text{Scrap Rate (\%)} = (\text{scrap_units} / \text{units_produced}) \times 100$$

4.11.2 11.2 Scrap Rate Thresholds

Scrap Rate	Classification	Required Action
<= 0.5%	Normal	Continue standard monitoring.
> 0.5% and <= 1.0%	Warning	Review scrap causes and affected components.
> 1.0%	Critical	Initiate documented investigation and quality escalation.

Scrap involving safety-critical components requires review regardless of the overall Scrap Rate.

4.12 12. Scrap Authorization

Scrap disposition should identify:

- affected batch;
- affected component or unit where applicable;
- reason for scrap;
- defect category;
- quantity;
- responsible authority;
- required traceability information.

Units designated as scrap must be physically or administratively controlled to prevent unintended return to production.

4.13 13. Relationship Between Defects, Rework, and Scrap

The following logical relationships should normally hold for production data:

```
rework_units <= defective_units
scrap_units <= defective_units
defective_units <= units_produced
```

Depending on the manufacturing process, rework and scrap classifications may represent different dispositions of defective units.

Analytical systems must validate these relationships before calculating quality KPIs.

Values violating basic production constraints must be treated as potential data-quality issues.

4.14 14. Data Validation

Before performing Rework Rate or Scrap Rate analysis, the underlying data should be evaluated for:

- duplicated batches;
- missing production quantities;
- invalid unit counts;
- negative values;
- inconsistent categories;
- anomalous numerical values.

For example:

```
units_produced = 80  
defective_units = 87
```

is logically invalid and must not be interpreted as an actual Defect Rate without validation.

When reliable correction is not possible, the affected observation should be flagged or excluded from the relevant analysis and the limitation documented.

4.15 15. Trend Analysis

Rework and scrap should be monitored over time rather than evaluated exclusively at individual batch level.

Trend analysis should consider:

- monthly Rework Rate;
- monthly Scrap Rate;
- production-line performance;
- vehicle-model performance;
- component categories;
- defect categories;
- supplier associations;
- production shifts.

A persistent increase may justify investigation even before a Critical threshold is exceeded.

4.16 16. Supplier-Related Rework and Scrap

When rework or scrap appears associated with supplied components, the analysis should consider:

- supplier identifier;
- component category;
- affected production volume;
- defect recurrence;
- total rework;
- total scrap;
- severity.

Significant supplier-related patterns must be evaluated according to the Supplier Quality Procedure.

Statistical association alone does not establish supplier responsibility.

4.17 17. Escalation Requirements

Formal escalation is required when:

- Rework Rate exceeds 3.0%;
- Scrap Rate exceeds 1.0%;
- repeated Warning conditions occur;
- rework repeatedly fails;
- scrap involves safety-critical components;
- significant supplier-related rework or scrap is identified;
- a recurring pattern indicates potential process loss of control.

Escalation must follow the Production Escalation Policy where applicable.

4.18 18. Responsibilities

4.18.1 Quality Engineer

Responsible for:

- evaluating rework eligibility;
- defining or verifying quality requirements;
- reviewing abnormal rework and scrap trends;
- initiating investigation when required;
- verifying reworked units where applicable.

4.18.2 Manufacturing Supervisor

Responsible for:

- coordinating approved rework activities;
- maintaining production traceability;
- preventing unauthorized disposition;
- supporting containment and investigation.

4.18.3 Supplier Quality Engineer

Responsible for:

- evaluating supplier-related rework and scrap;
- coordinating supplier corrective actions where required.

4.18.4 Quality Manager

Responsible for:

- reviewing Critical rework or scrap conditions;
 - approving significant quality dispositions where required;
 - supporting escalation decisions.
-

4.19 19. Use of Maranello AI

Maranello AI may support rework and scrap management by:

- calculating Rework Rate;
- calculating Scrap Rate;
- comparing rates across production lines;
- identifying recurring patterns;
- analyzing trends over time;
- identifying supplier or component associations;
- retrieving applicable rework and scrap requirements;

- highlighting conditions exceeding documented thresholds.

For example, the system may calculate that a production line has a Rework Rate of 3.4% and retrieve this procedure to determine that the condition is classified as Critical.

Maranello AI provides analytical decision support and does not autonomously authorize physical rework, scrap disposition, or release of affected production.

4.20 20. Records and Traceability

Rework and scrap activities should preserve sufficient information to reconstruct:

- what happened;
- which production was affected;
- why the disposition was selected;
- what corrective activity was performed;
- how conformity was verified;
- who authorized the relevant decision.

Traceability must be maintained throughout the lifecycle of the affected production.

4.21 21. Related Documents

This procedure should be interpreted together with:

- Manufacturing Quality Policy;
- Non-Conformity Management Procedure;
- Supplier Quality Procedure;
- Production Escalation Policy.

4.22 22. Document Control

Field	Value
Document ID	RSP-001
Version	1.0
Owner	Quality & Manufacturing Operations
Classification	Internal
Effective Date	2025-01-01
Review Cycle	Annual
Status	Active

5 Production Escalation Policy

Document ID: PEP-001

Version: 1.0

Owner: Quality & Manufacturing Operations

Classification: Internal

Effective Date: 2025-01-01

Review Cycle: Annual

Status: Active

5.1 1. Purpose

This policy defines the escalation framework used to manage significant manufacturing, quality, supplier, and production-line events.

The objective is to ensure that abnormal conditions are escalated consistently according to their severity, operational impact, recurrence, and potential risk.

The policy establishes escalation levels, triggering conditions, responsibilities, notification requirements, and criteria for returning to normal operations.

5.2 2. Scope

This policy applies to events associated with:

- manufacturing quality;
- production-line performance;
- production downtime;
- non-conforming batches;
- rework and scrap;
- supplier-related quality issues;
- recurring manufacturing problems;
- safety-critical defects;
- abnormal quality KPI performance.

It applies to all production lines, shifts, vehicle models, manufacturing teams, and suppliers included within the Quality & Manufacturing Operations scope.

5.3 3. Escalation Principles

Production escalation shall follow these principles:

1. escalation must be proportional to the severity and potential impact of the event;
 2. product quality and safety take priority over production continuity;
 3. Critical conditions require timely management visibility;
 4. repeated Warning conditions may justify escalation even when individual events remain below Critical thresholds;
 5. operational decisions must be based on validated information;
 6. containment does not replace root-cause analysis;
 7. escalation decisions and actions must remain traceable;
 8. production may return to normal operation only after applicable risks have been adequately controlled.
-

5.4 4. Escalation Levels

Maranello AI Quality & Manufacturing Operations uses three operational escalation levels.

Level	Classification	Description
Level 1	Attention	Local deviation requiring increased monitoring or operational review.
Level 2	Significant	Confirmed quality or operational issue requiring structured investigation and cross-functional involvement.
Level 3	Critical	Severe condition requiring immediate containment, management notification, and formal escalation.

The escalation level should reflect the most severe applicable condition.

5.5 5. Level 1 — Attention

Level 1 applies when a deviation requires additional monitoring but does not represent an immediate significant manufacturing risk.

Typical Level 1 triggers include:

- Defect Rate above 2.0% and up to 3.5%;
- Rework Rate above 1.5% and up to 3.0%;
- Scrap Rate above 0.5% and up to 1.0%;
- Quality Score between 85 and below 90;
- production-line downtime above 45 and up to 90 minutes;
- isolated non-critical quality deviation;
- emerging negative trend requiring observation.

5.5.1 Required Actions

Level 1 actions may include:

- notify the responsible Quality Engineer;
- increase monitoring;
- review subsequent production;
- evaluate the dominant defect category;
- verify production conditions;
- document relevant observations.

Production may normally continue unless additional evidence indicates greater risk.

5.6 6. Level 2 — Significant

Level 2 applies when a confirmed manufacturing or quality issue requires structured investigation or coordinated action.

Typical Level 2 triggers include:

- repeated Level 1 conditions;
- three consecutive Warning periods for the same relevant KPI;

- recurring non-conformities;
- formal supplier quality review;
- repeated abnormal rework;
- significant scrap pattern;
- failed corrective action;
- repeated production disruption;
- a Major non-conformity.

5.6.1 Required Actions

Level 2 requires:

1. notification of the Quality Engineer;
2. notification of the Manufacturing Supervisor;
3. documented containment where appropriate;
4. structured investigation;
5. assignment of responsible stakeholders;
6. root-cause analysis where required;
7. definition of corrective actions;
8. verification of corrective-action effectiveness.

The Quality Manager should be informed when the event has significant cross-functional impact.

5.7 7. Level 3 — Critical

Level 3 represents the highest escalation level.

It applies when a condition may create significant product, customer, safety, manufacturing, or business risk.

Typical Level 3 triggers include:

- Defect Rate above 3.5%;
- Rework Rate above 3.0%;
- Scrap Rate above 1.0%;
- Quality Score below 85;
- production-line downtime above 90 minutes when operationally significant;
- production-line downtime above 180 minutes;
- safety-critical defect;
- Critical non-conformity;
- Supplier Defect Rate above 4.0% over a validated monitoring period;
- repeated unresolved Major non-conformities;
- confirmed loss of manufacturing process control;
- ineffective corrective actions associated with significant risk.

5.7.1 Required Actions

Level 3 requires:

1. immediate Quality notification;
2. immediate Manufacturing Operations notification;
3. Quality Manager notification;
4. containment of affected production where applicable;
5. identification and isolation of affected batches;
6. formal investigation;
7. root-cause analysis;
8. documented corrective actions;
9. verification before return to normal operation.

Additional management or supplier involvement may be required depending on the event.

5.8 8. Production Hold

A production hold may be required when continued production could increase quality, safety, or operational risk.

Conditions that may justify a production hold include:

- safety-critical defects;
- confirmed severe loss of process control;
- inability to determine the extent of affected production;
- repeated Critical defects;
- ineffective containment;
- inability to verify product conformity.

A Critical KPI threshold alone does not automatically require a complete production shutdown.

The Quality and Manufacturing teams must evaluate the specific risk and determine the appropriate containment response.

5.9 9. Downtime Escalation

Production-line downtime is classified as follows:

Downtime	Classification	Escalation
<= 45 minutes	Normal	No escalation required.
> 45 and <= 90 minutes	Warning	Level 1 operational review.
> 90 and <= 180 minutes	Critical	Level 3 escalation when validated and operationally significant.
> 180 minutes	Severe Critical	Immediate Level 3 escalation and management notification.

Before escalation, implausible downtime values must be checked for potential data-quality errors.

For example, an unusually large recorded downtime value should not automatically trigger operational action if the underlying record is demonstrably invalid.

5.10 10. Quality KPI Escalation Matrix

KPI	Normal	Warning	Critical
Defect Rate	<= 2.0%	> 2.0% and <= 3.5%	> 3.5%
Rework Rate	<= 1.5%	> 1.5% and <= 3.0%	> 3.0%
Scrap Rate	<= 0.5%	> 0.5% and <= 1.0%	> 1.0%
Quality Score	>= 90	>= 85 and < 90	< 85
Downtime	<= 45 min	> 45 and <= 90 min	> 90 min

The Manufacturing Quality Policy remains the authoritative source for KPI definitions and detailed quality requirements.

5.11 11. Recurring Warning Conditions

Repeated Warning conditions must not be treated indefinitely as isolated events.

Escalation to Level 2 is required when:

- the same KPI remains in Warning status for three consecutive monitored periods;
- the same significant defect category repeatedly affects the same production scope;
- the same production line repeatedly enters Warning status;
- corrective actions do not produce sustained improvement.

Recurring Warning conditions may be escalated to Level 3 when evidence indicates increasing severity or loss of process control.

5.12 12. Supplier Escalation

Supplier-related escalation must consider both manufacturing impact and supplier performance.

Level 2 supplier escalation may be appropriate when:

- Supplier Defect Rate enters Warning status;
- recurring supplier-related defects are identified;
- formal supplier quality review is required;
- supplier corrective actions are ineffective.

Level 3 supplier escalation is required when:

- Supplier Defect Rate exceeds 4.0% over a validated monitoring period;
- a supplied safety-critical component presents a confirmed quality issue;
- supplier-related defects create significant production or customer risk;
- repeated supplier corrective actions fail.

Detailed supplier requirements are defined in the Supplier Quality Procedure.

5.13 13. Non-Conformity Escalation

Confirmed non-conformities must be escalated according to their classification.

5.13.1 Minor Non-Conformity

Normally managed locally unless recurrence indicates a systemic issue.

5.13.2 Major Non-Conformity

Normally requires Level 2 escalation.

5.13.3 Critical Non-Conformity

Requires Level 3 escalation.

The Non-Conformity Management Procedure defines the detailed investigation and closure process.

5.14 14. Data Validation Before Escalation

Automated or analytical escalation decisions must use validated data.

Before using a KPI as the basis for escalation, relevant data should be evaluated for:

- duplicates;
- missing values;
- invalid quantities;
- inconsistent categories;
- inconsistent dates;
- extreme outliers;
- logical inconsistencies.

For example:

```
units_produced = 70
defective_units = 78
```

must not be interpreted as a valid Critical Defect Rate.

The record must first be investigated as a potential data-quality problem.

When uncertainty remains, Maranello AI should identify the condition as requiring data validation rather than automatically presenting it as a confirmed operational escalation.

5.15 15. Escalation Communication

An escalation communication should contain sufficient information for stakeholders to understand the event.

Where available, it should include:

- affected production scope;
- relevant production line;
- affected batch or batches;
- date or monitoring period;
- relevant KPI;
- measured value;
- applicable threshold;
- defect category;
- supplier or component association;
- escalation level;
- immediate containment;
- responsible stakeholders;
- next required action.

5.16 16. Responsibilities

5.16.1 Quality Engineer

Responsible for:

- validating quality-related escalation triggers;
- evaluating affected production;
- initiating non-conformity activities where required;
- supporting investigation and corrective actions.

5.16.2 Manufacturing Supervisor

Responsible for:

- coordinating operational response;
- implementing containment;
- controlling affected production;

- supporting investigation.

5.16.3 Supplier Quality Engineer

Responsible for:

- managing supplier-related escalation;
- coordinating supplier containment and corrective actions;
- monitoring supplier recovery.

5.16.4 Quality Manager

Responsible for:

- reviewing Level 3 quality events;
 - coordinating significant cross-functional escalation;
 - approving major quality decisions where required;
 - ensuring appropriate follow-up.
-

5.17 17. Return to Normal Operation

An escalated condition may return to normal monitoring only when applicable requirements have been satisfied.

Depending on severity, this may require:

- successful containment;
- validated production data;
- completed investigation;
- implemented corrective actions;
- verified effectiveness;
- acceptable subsequent production;
- appropriate Quality approval.

A temporary improvement in one production batch is not necessarily sufficient to close a recurring or Critical issue.

5.18 18. Use of Maranello AI

Maranello AI may support escalation management by:

- calculating manufacturing KPIs;
- identifying values exceeding documented thresholds;
- detecting recurring patterns;
- analyzing historical production performance;
- retrieving applicable escalation rules;
- identifying potentially affected suppliers, lines, shifts, or components;
- generating evidence-based escalation summaries.

For example:

User question:

"What was the defect rate for Line 3 last month,
and does it require escalation?"

Maranello AI may:

1. use the Data Agent to clean and analyze the production dataset;
2. calculate the validated Defect Rate for Line 3;

3. use the RAG system to retrieve the applicable threshold;
4. compare the calculated result with this policy;
5. explain the applicable escalation level and required action.

Maranello AI provides decision support and must not independently authorize:

- production shutdown;
 - production restart;
 - release of isolated batches;
 - closure of Critical non-conformities;
 - final safety-critical decisions.
-

5.19 19. Escalation Example

Consider a validated monthly analysis with the following result:

Production Line: Line 3

Defect Rate: 3.8%

The Manufacturing Quality Policy defines a Defect Rate above 3.5% as Critical.

According to this Production Escalation Policy, the condition therefore requires Level 3 escalation.

The expected response includes:

- Quality notification;
- Manufacturing Operations notification;
- Quality Manager notification;
- evaluation of affected production;
- appropriate containment;
- formal investigation;
- corrective-action planning.

The analytical result alone does not determine whether a complete production shutdown is necessary.

That decision requires evaluation of the specific operational and quality risk.

5.20 20. Related Documents

This policy should be interpreted together with:

- Manufacturing Quality Policy;
 - Non-Conformity Management Procedure;
 - Supplier Quality Procedure;
 - Rework and Scrap Procedure.
-

5.21 21. Document Control

Field	Value
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